

The Complete Journey: From $R^2 = -0.46$ to 0.05122

Sources

- User-provided content — complete journey narrative, mathematical explanations, and analysis

Core Thesis

Disciplined, evidence-driven data science workflow that systematically diagnosed a contaminated dataset problem, avoided the trimming trap, and achieved a good score by optimizing for stability over best-case fit.

The Journey at a Glance

Visual progression showing the four stages:

- -0.46 (Naive) $\rightarrow -0.093$ (OLS) $\rightarrow 0.04875$ (Ridge) $\rightarrow 0.05122$ (Tuned + Feature Drop)
Brief one-line description of each stage

Problem Statement

Problem: Predict a continuous target from 15 anonymized features (R^2 metric).

Investigation & Discovery

- Naive linear regression scored $R^2 = -0.46$ — something was wrong
- Discovered: target = weak linear signal + $\sim 3\%$ corrupted outliers ($\pm 40,000$)
- Systematically tested and rejected: product, ratio, polynomial, and tree-based hypotheses
- Conclusion: outliers are irreducible noise, unpredictable by any model

The Critical Trap: Why Trimming Fails

- Trimming outliers gave CV R^2 of 0.95 — but honest train/val split revealed only ~ 0.005 on full data

- Why: R^2 is dominated by outliers (one outlier contributes ~1.8 MILLION times more than one normal row)
- Even though outliers are ~3% of rows, they account for ~95%+ of total variance in denominator
- The mismatch: optimized on "clean world" but graded in "real world"

Include the R^2 formula and the analogy:

- 97 questions worth 1 point each (clean rows)
- 3 questions worth 1,000,000 points each (outliers)

The Proof

Model A — R^2 on CLEAN val : 0.9424	← what trimming "showed" us
Model A — R^2 on FULL val : 0.0053	← what Kaggle actually grades

Same model, same predictions — only difference is whether evaluation set contains outliers.

Implemented Strategy: Optimize for Stability

- Trained Ridge regression on full dataset (including outliers)
- Heavy regularization ($\alpha=4000$) to capture reliable linear signal while resisting high leverage of outliers
- Validated via 40-seed stability test (mean +0.041, never negative)
- Dropped 3 zero-coefficient noise features (x11, x12, x13) for final marginal gain
- Result: 0.05122

Summary Comparison Table

Aspect	Trimmed (Trap)	Full Data (Correct)
Optimizes	Clean core (5% of variance)	Full distribution (incl. 95% from outliers)
CV R^2	0.95 (misleading)	~0.04 (honest)

Aspect	Trimmed (Trap)	Full Data (Correct)
Kaggle R^2	0.005 ❌	0.051 ✅
Why	Train/eval mismatch distribution	Matches the real metric

Key Takeaway

When an evaluation metric is dominated by unpredictable noise, the goal shifts from maximizing fit to maximizing stability and expected performance. Regularization and noise-feature removal beat complex models on contaminated data.

The General Lesson

Never clean your data in a way that doesn't match how you'll be evaluated. If your test set / production data will contain outliers, and your metric is sensitive to them, then removing them from training/validation gives you a falsely optimistic score on a distribution that won't exist at evaluation time.

What Was Done Right (Final Page)

- ✅ Explored before modeling
- ✅ Formed and tested hypotheses (didn't assume)
- ✅ Caught the CV-vs-reality trap with an honest holdout
- ✅ Used limited submissions strategically
- ✅ Reached the realistic ceiling with a simple, defensible model

Final score: **0.05122** 🎯

Link to code - https://github.com/manish0338/ml-hw-Spring-2026-python/blob/main/machine_learning_project.ipynb